Temporal Data Lineage and Impact Analysis in Oracle Database

Evolution to Oracle Graph Architecture with Time-Travel Capabilities

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Agenda

Intro to SQL Property Graph
(aka Operational Property Graphs)

Presented by: Karin

A Real-World SQL Property Graph Use Case implemented using Oracle Application Express

Presented by: Tomislav

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SQL (Operational) Property Graphs
A Brief Intro

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ORACLE AI Database 26ai

Data Types

Vector, Spatial, JSON, Text, XML, Relational

Workloads

AI / ML, IoT, Analytical, Streaming, OLTP, Graph, Blockchain, Distributed

Tools

APEX, ORDS, SQLcl, MCP Server

AutoML, Enterprise Manager, SQL Developer VS Code, Agent Factory, Private AI Container

Dev Interfaces

SQL First, RESTful, MongoDB API

Language Support: PL, Java, RUST, C++, NODE.JS, .NET, JS, RUBY, C, PL/SQL, PHP, GO

Runs Anywhere

On-Premises, Containers, Oracle Cloud, Autonomous AI Database, Cloud@Customer, @Azure / @AWS / @GCP

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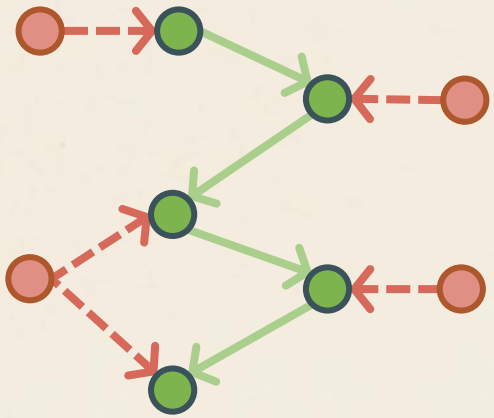
The (Labeled) Property Graph Data Model



A representation of the real world, focusing on relationships using

- Vertices (aka Nodes) representing a *Thing*
- Edges representing connections between vertices
- Properties expressed as key-value pairs attached to vertices and/or edges
- Labels representing the type of a vertex or edge; one or more labels can be applied

Graph-based queries:

- Fast traversals
- Path finding
- Identify patterns
- Extract subgraphs

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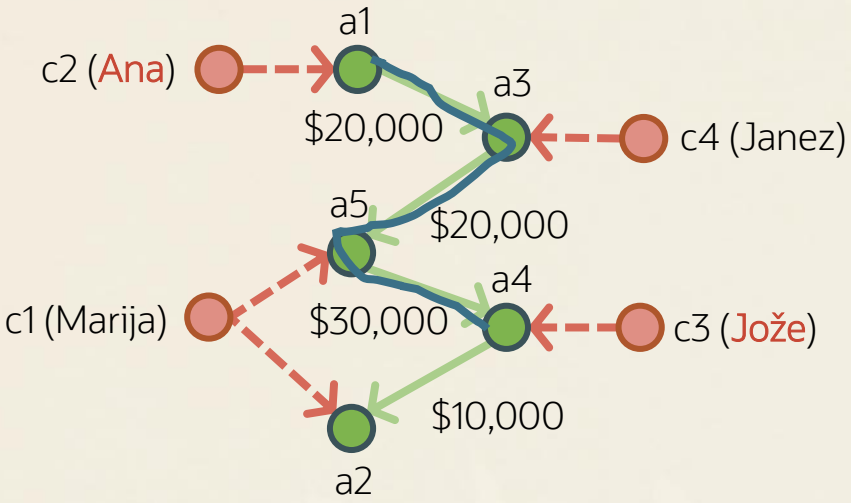
Find Paths in Property Graphs

*Is there **any money flow between Ana & Jože**?*



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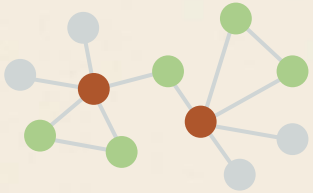
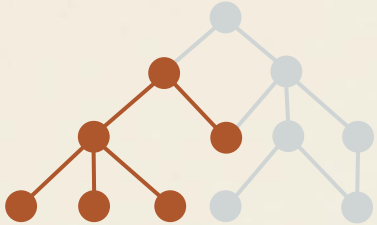
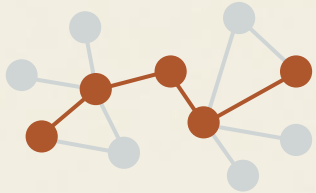
Typical Patterns in Property Graphs

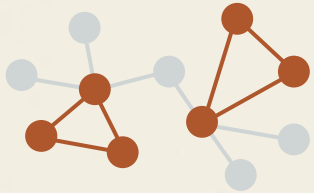
Networks and hierarchies





Paths and patterns





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Two Steps: Create and Analyze a Property Graph using SQL

Prerequisite: Oracle Database 23ai or higher

- Build and query Property Graphs using SQL/PGQ
- SQL Standard Extension ☞ SQL:2023 (ISO/IEC 9075-16)

Step 1: Create a graph

CREATE PROPERTY GRAPH to build the graph as a metadata object over the original data

- No data copy
- No data sync ☞ Operational Property Graph

Step 2: Query the graph

MATCH clause inside the GRAPH_TABLE function to run Pattern-Matching Queries

Schema flexibility through JSON

- JSON document as a property
- JSON element as a property

Combines all converged DB features using SQL

See also:
www.oracle.com/a/ocom/docs/database/operational_property_graph_with_23c.pdf
blogs.oracle.com/database/post/property-graphs-in-oracle-database-26ai-the-sql-pgq-standard

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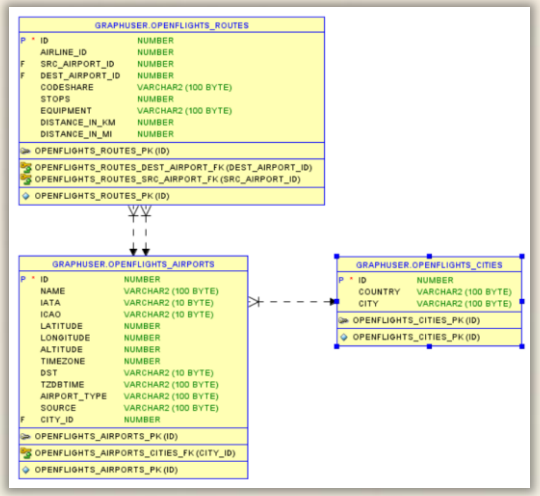
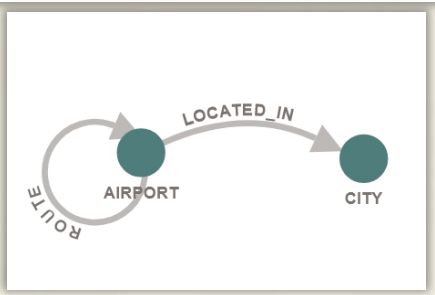
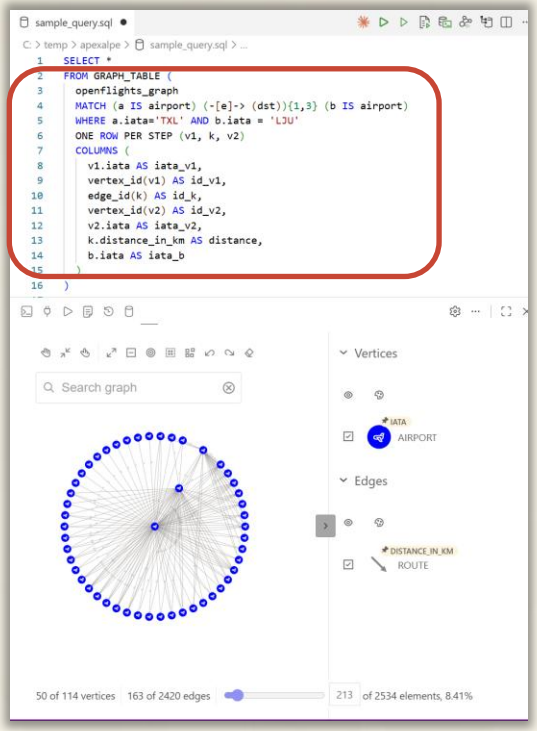
From Relational/JSON to Labeled Property Graph




```
CREATE PROPERTY GRAPH IF NOT EXISTS openflights_graph
  VERTEX TABLES (
    openflights_airports AS airports
      KEY ( id )
      LABEL airport
      PROPERTIES ARE ALL COLUMNS,
    openflights_cities AS cities
      KEY ( id )
      LABEL city
      PROPERTIES ARE ALL COLUMNS
  )
  EDGE TABLES (
    openflights_routes AS routes
      SOURCE KEY ( src_airport_id ) REFERENCES airports (id)
      DESTINATION KEY ( dest_airport_id ) REFERENCES airports (id)
      LABEL route
      PROPERTIES ARE ALL COLUMNS,
    openflights_airports AS airports_in_cities
      SOURCE KEY ( id ) REFERENCES airports (id)
      DESTINATION KEY ( city_id ) REFERENCES cities (id)
      LABEL located_in
      NO PROPERTIES
  )
```





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Tools to work with SQL Property Graphs in 26ai

Create and query graphs

- SQL Developer
- SQL Developer Extension for VS Code
- Graph Studio for Autonomous Database
- SQLcl
- APEX (Plug-in, Sample Graph Viz app)
- Any other SQL client

Visualize graphs

- Graph Visualization JavaScript Library
  - Embeddable in any web application
- Used by
  - SQL Developer Extension for VS Code
  - APEX Graph Visualization Plug-in
  - Graph Studio for Autonomous Database
- 3rd Party:
  - Linkurious OGMA

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Social Commerce
ORACLE 26AI DEMO

- Welcome
- Schema & Data
- Dashboard
- Social Vector Trends
- Influencer Graph
- Fulfillment Map
- Orders
- OML Analytics
- Ask Your Data
- Agent Console

JC **Jessica Chen**
Admin

Converged Database
Relational · JSON · Graph · Vector · Spatial · OML

Oracle AI Database 26ai — Converged Architecture Demo

One Database. Seven Capabilities. Zero Complexity.

Most companies run **7+ separate systems** for relational data, document stores, graph databases, vector search, spatial, ML platforms, and AI agents. Each one adds latency, sync failures, and operational burden.

Oracle AI Database 26ai runs all seven natively — in a single engine, on the same data, with ACID guarantees across every query.

Live Architecture

Browser Client
React 18 · Vite · Recharts · Leaflet

↓ HTTPS · Port 3001

OCI Compute Instance — 4 OCPU, 24 GB RAM
VM.Standard.A1.Flex — Oracle Linux 9

- **Node.js 20** — Express API
- **PM2** — Process Mgr

↓ Oracle Thin Driver · mTLS Wallet

Autonomous Database — 4 ECPU, 20 GB
Oracle AI Database 26ai — Always Free

AI-Assisted Development

Built entirely with AI-powered tooling

VS Code
Developer IDE

- **Oracle Code Assist** — Context-aware · PL/SQL + JS · Oracle-optimized
- **SQLcl MCP Server** — Schema introspection · SQL execution · DDL ops
- **LLM** — Multi-step reasoning · Full-stack generation

VS Code › Code Assist + SQLcl MCP › LLM › Oracle ADB

What This Demo Solves

The Problem (Traditional Stack)

- ✗ PostgreSQL for orders + MongoDB for social posts
- ✗ Neo4j for influencer graphs + Pinecone for vectors
- ✗ PostGIS for spatial + separate ML platform
- ✗ Data sync across 6 systems = eventual consistency bugs
- ✗ Each system needs its own DevOps, backups, scaling

The Solution (Oracle AI Database 26ai)

- ✓ One database engine — relational, JSON, graph, vector, spatial, ML, AI agents
- ✓ Single ACID transaction spans all data models
- ✓ Vector search with ONNX models loaded into the DB kernel
- ✓ SDO_GEOMETRY spatial queries on the same tables as orders
- ✓ Select AI — natural language to SQL with zero application code
- ✓ One backup, one HA config, one security perimeter

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Summary

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Graphs offer a different view on your data.

With Oracle Database 26ai, you can use your existing tools to work with graphs.

Oracle Graph is the enterprise-grade graph database.

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More information

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Useful Resources

General information

- www.oracle.com/database/graph/
- Graph Size Estimator www.oracle.com/webfolder/assets/graph-size-estimator/index.html
- PGQL Spec 2.1 pgql-lang.org/spec/2.1/

Documentation Release 26.1

- Oracle Property Graph docs.oracle.com/en/database/oracle/property-graph/26.1/index.html
- Python API docs.oracle.com/en/database/oracle/property-graph/26.1/pypgx/pypgx/api.html
- Java API https://docs.oracle.com/en/database/oracle/property-graph/26.1/spgjv/index.html
- Graph Server REST API docs.oracle.com/en/database/oracle/property-graph/26.1/spgdg/graph-server-rest-api-version-2.html

GitHub

- Machine Learning Workshop with Oracle Property Graph github.com/operard/mlgraph
- github.com/oracle-samples/oracle-graph

Oracle Cloud MarketPlace

- Graph Server and Client cloud.oracle.com/marketplace/application/74883066

YouTube

- youtube.com/@oraclespatialandoraclegraph
- Analytics and Data Techcasts: youtube.com/@analyticsanddatatechcasts90

Blogs & Posts

- Oracle Database Insider – Graph: blogs.oracle.com/database/category/db-graph
- Medium: medium.com/tag/oracle-graph
- Database & Cloud Technology: blogs.oracle.com/coretec/
- *Oracle AI Database from the ground up*: jasperan.github.io/visual-oracledb/#graphs

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Links for Downloads and Documentation

- SQL Developer
  oracle.com/database/sqldeveloper/technologies/download/
- SQLcl
  oracle.com/database/sqldeveloper/technologies/sqlcl/download/
- Graph Server and Clients including Graph Visualization Library
  oracle.com/database/technologies/spatialandgraph/property-graph-features/graph-server-and-client/graph-server-and-client-downloads.html
- APEX Sample Graph App
  github.com/oracle/apex/tree/24.2/sample-apps/sample-graph-visualizations
- Graph Developer´s Guide for Property Graph – Graph Visualization
  docs.oracle.com/en/database/oracle/property-graph/26.1/spgdg/graph-visualization-application.html
- Oracle Graph Visualization Library JavaScript API
  docs.oracle.com/en/database/oracle/property-graph/26.1/pgjsd/index.html
- Get Started with APEX Graph Visualization Plug-in
  docs.oracle.com/en/cloud/paas/autonomous-database/csgru/get-started-apex-graph-visualization-plug.html

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Thank You !

For more information on Oracle Graph
go to oracle.com/goto/graph

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